Sahaj Bhadja

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Summary

Software engineer with 2+ years building **performance-critical backend systems** and infrastructure. Proficient in Python and C++ fundamentals, with hands-on experience in containerization (Docker), Linux environments, and cloud infrastructure optimization.

Education

University of Southern California, Los Angeles, CA — **August 2024 – May 2026**
Master of Science in Computer Science — *GPA: 3.75/4.0*
Coursework: Operating Systems, Distributed Systems, Analysis of Algorithms, Computer Networks

Pandit Deendayal Energy University, Gandhinagar, India — **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering — *GPA: 3.88/4.0*
Coursework: Systems Programming (C/C++), Operating Systems, Data Structures, Computer Networks

Experience

USC Dornsife — **Los Angeles, CA**
Software Engineer — *September 2025 – Present*
- Developed **performance-critical backend services** in Python, implementing monitoring and telemetry systems to track system health, resource utilization, and identify bottlenecks across production infrastructure.
- Benchmarked and profiled application performance, **eliminating bottlenecks** through systematic optimization to achieve 99%+ reliability for services processing 5K+ monthly workloads.

Grad8 — **Los Angeles, CA**
Software Engineer Intern — *June 2025 – August 2025*
- Built **daemon-like backend services** in Python handling 10K+ daily requests, implementing process orchestration and resource management for production AI workloads with high availability requirements.
- Profiled and optimized **system performance**, troubleshooting infrastructure issues and implementing fixes to improve reliability across containerized deployment environments.

USC AD Labs — **Los Angeles, CA**
ML Research Assistant — *January 2025 – May 2025*
- Optimized **compute-intensive workloads** on GPU infrastructure, improving processing efficiency by 18% through algorithmic optimization and resource management strategies.
- Developed Python-based **data processing pipelines** on Linux systems, working with cloud compute resources and implementing performance profiling for ML training workflows.

Aasma Technology Solutions — **India**
Software Developer Engineer - (SecureSync) — *August 2023 – July 2024*
- Designed and implemented **backend infrastructure systems** in Python, improving system efficiency by **68%** through performance optimization, caching strategies, and resource management.
- Secured and hardened **cloud infrastructure** following security best practices, implementing access controls and monitoring across multi-tenant deployment environments.

Aasma Technology Solutions — **India**
Full Stack Developer Intern - (VibrantChem) — *May 2023 – August 2023*
- Deployed **containerized applications using Docker** on Linux environments, configuring container orchestration and troubleshooting deployment issues across development and production systems.
- Built backend services handling **2M+ records with real-time processing**, implementing WebSocket connections and optimizing database queries for low-latency performance.

Skills

Languages: Python (primary), C++ (fundamentals), C, JavaScript, SQL, Bash scripting
Systems & Infrastructure: Linux, Docker, containerization, cloud infrastructure (AWS, GCP)
Performance: Profiling, benchmarking, optimization, monitoring, telemetry
Tools: Git, CI/CD pipelines, process management, resource orchestration

Projects

Linguistix Code | *Python, Docker, AWS (ECS, Fargate), Linux*
- Built **containerized application infrastructure** with Docker deployment on Linux, implementing automated orchestration pipelines and resource management for scalable cloud compute.

Big-Brain Live | *Python, PostgreSQL, FAISS, Performance Optimization*
- Developed **performance-optimized backend** processing 50K+ documents, implementing algorithmic optimizations and benchmarking to achieve 95% query accuracy with sub-second latency.

FinNexus Live | *TypeScript, MongoDB, WebSockets, Real-time Processing*
- Built **high-throughput backend services** supporting 50K+ users and 1M+ transactions, implementing real-time data synchronization and connection management for reliable operations.